

Dynamical restrictions on positivity of tangent bundles

(Joint work with Sheng Meng)

Workshop on birational geometry, Iskovskikh Seminar

Guolei ZHONG

Institute for Basic Science, Center for Complex Geometry

5 December 2024

Contents

1. Background & Terminology
2. Previous results
3. Main results
4. Application: the Kawaguchi-Silverman conjecture

1. Background & Terminology

Motivation – Classify algebraic varieties

- Positivity of tangent sheaves & Algebraic dynamics;
- Both of them share similar fundamental building blocks: abelian varieties and rationally connected varieties.
- Our guiding expectation: **Dynamics $\implies$ Positivity $\implies$ Geometry.**

1. Background & Terminology

Motivation – Classify algebraic varieties

- Positivity of tangent sheaves & Algebraic dynamics;
 - Both of them share similar fundamental building blocks: abelian varieties and rationally connected varieties.
 - Our guiding expectation: **Dynamics $\implies$ Positivity $\implies$ Geometry.**
-
- We work over the field $k = \bar{k}$ of char $k = 0$. For a projective variety X , we denote by $N^1(X) = \text{NS}(X) \otimes_{\mathbb{Z}} \mathbb{R}$ the Néron-Severi $\mathbb{R}$ -space.
 - The **tangent sheaf** $\mathcal{T}_X$ of a normal variety X is defined as the dual of its sheaf of Kähler differentials Ω_X .
 - Let $D \geq 0$ be a reduced divisor on a normal variety X , and U a big open subset such that $(U, D|_U)$ is log smooth. Let $\Omega_X^{[1]}(\log D)$ be the reflexive sheaf on X whose restriction to U is the sheaf of the usual logarithmic differential 1-form. Its dual $\mathcal{T}_X(-\log D)$ is called the **logarithmic tangent sheaf**.

1. Background & Terminology (Positivity)

Definition (Positivity of reflexive sheaves)

Let $\mathcal{E}$ be a torsion-free coherent sheaf on a normal projective variety X .

- $\mathcal{E}$ is **pseudo-effective** (psef) if $\forall a \in \mathbb{Z}_{>0}$ and $\forall$ ample divisor A , $\exists b \in \mathbb{Z}_{>0}$ such that $\text{Sym}^{[ab]} \mathcal{E} \otimes \mathcal{O}_X(bA)$ is globally generated at a general point (Viehweg, 1983), (Nakayama, 2004); $\Rightarrow \mathcal{O}_{\mathbb{P}_X(\mathcal{E})}(1)$ psef.
- $\mathcal{E}$ is **ample** (resp. **nef**, **big**) if the tautological line bundle $\mathcal{O}_{\mathbb{P}_X(\mathcal{E})}(1)$ is ample (resp. nef, big) on the Grothendieck projectivization $\mathbb{P}_X(\mathcal{E})$.
- $\mathcal{E}$ is **numerically flat** if both $\mathcal{E}$ and its dual $\mathcal{E}^\vee$ are nef and locally free.

1. Background & Terminology (Positivity)

Definition (Positivity of reflexive sheaves)

Let $\mathcal{E}$ be a torsion-free coherent sheaf on a normal projective variety X .

- $\mathcal{E}$ is **pseudo-effective** (psef) if $\forall a \in \mathbb{Z}_{>0}$ and $\forall$ ample divisor A , $\exists b \in \mathbb{Z}_{>0}$ such that $\text{Sym}^{[ab]} \mathcal{E} \otimes \mathcal{O}_X(bA)$ is globally generated at a general point (Viehweg, 1983), (Nakayama, 2004); $\Rightarrow \mathcal{O}_{\mathbb{P}_X(\mathcal{E})}(1)$ psef.
- $\mathcal{E}$ is **ample** (resp. **nef**, **big**) if the tautological line bundle $\mathcal{O}_{\mathbb{P}_X(\mathcal{E})}(1)$ is ample (resp. nef, big) on the Grothendieck projectivization $\mathbb{P}_X(\mathcal{E})$.
- $\mathcal{E}$ is **numerically flat** if both $\mathcal{E}$ and its dual $\mathcal{E}^\vee$ are nef and locally free.

Example

- $\mathcal{E} := \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$ is not psef, but $\mathcal{O}_{\mathbb{P}(\mathcal{E})}(1)$ is effective.
- (Iwai-Matsumura-Z., 2023) Let X be a blow-up of $\mathbb{P}^1 \times \mathbb{P}^1$ along n distinct points lying in the same fibre of the first projection. Then $H^0(X, \mathcal{T}_X) \neq 0$. However, $\mathcal{T}_X$ is not psef whenever $n \geq 5$.

1. Background & Terminology (Dynamics)

To extend the notion of non-isomorphic surjective endomorphisms of a projective variety of Picard number 1 to the higher Picard number case, we recall the following definition.

Definition (Int-amplified endomorphisms) (Meng, 2020)

A surjective endomorphism $f: X \rightarrow X$ of a projective variety X is called **int-amplified**, if the linear operation $f^*|_{N^1(X)}$ has all e.vs. of modulus > 1
 $\iff f^*H - H$ is ample for some ample divisor $H \iff$ the topological degree is strictly larger than the other dynamical degrees.

1. Background & Terminology (Dynamics)

To extend the notion of non-isomorphic surjective endomorphisms of a projective variety of Picard number 1 to the higher Picard number case, we recall the following definition.

Definition (Int-amplified endomorphisms) (Meng, 2020)

A surjective endomorphism $f: X \rightarrow X$ of a projective variety X is called **int-amplified**, if the linear operation $f^*|_{N^1(X)}$ has all e.vs. of modulus > 1
 $\iff f^*H - H$ is ample for some ample divisor $H \iff$ the topological degree is strictly larger than the other dynamical degrees.

Example

- A normal toric pair has a trivial logarithmic tangent sheaf, has psef tangent sheaf, and always admits an int-amplified endomorphism.
- The product $f \times g$ of endomorphisms on a product variety $X \times Y$ is int-amplified if and only if both f and g are int-amplified.

2. Previous results (Positivity $\Rightarrow$ Geometry)

Theorem (Mori, 1979)=Hartshorne conjecture; cf. (Siu-Yau, 1980)

Every irreducible n -dimensional non-singular projective variety with **ample** tangent bundle defined over a field $k = \bar{k}$ of char $k = 0$ is $\mathbb{P}_k^n$.

2. Previous results (Positivity $\Rightarrow$ Geometry)

Theorem (Mori, 1979)=Hartshorne conjecture; cf. (Siu-Yau, 1980)

Every irreducible n -dimensional non-singular projective variety with **ample** tangent bundle defined over a field $k = \bar{k}$ of char $k = 0$ is $\mathbb{P}_k^n$.

Theorem (Demailly-Peternell-Schneider, 1994), (Cao, 2013)

A compact Kähler manifold with **nef** tangent bundle, up to a finite étale cover, admits a locally trivial fibration over a complex torus whose fibre is a Fano manifold with nef tangent bundle.

(Wang, 2024) recently proved that the tangent bundle of a fibre is big.

2. Previous results (Positivity $\Rightarrow$ Geometry)

Theorem (Mori, 1979)=Hartshorne conjecture; cf. (Siu-Yau, 1980)

Every irreducible n -dimensional non-singular projective variety with **ample** tangent bundle defined over a field $k = \bar{k}$ of char $k = 0$ is $\mathbb{P}_k^n$.

Theorem (Demailly-Peternell-Schneider, 1994), (Cao, 2013)

A compact Kähler manifold with **nef** tangent bundle, up to a finite étale cover, admits a locally trivial fibration over a complex torus whose fibre is a Fano manifold with nef tangent bundle.

(Wang, 2024) recently proved that the tangent bundle of a fibre is big.

Theorem (Höring-Liu-Shao, 2022)

A del Pezzo surface of deg d has big (resp. psef) tangent bundle $\Leftrightarrow d \geq 5$ (resp. $d \geq 4$); a smooth hypersurface of dimension ≥ 3 and of degree d has big or psef tangent bundle if and only if $d \leq 2$.

2. Previous results (Positivity $\Rightarrow$ Geometry)

Theorem (Hosono-Iwai-Matsumura, 2022)

A complex projective manifold with **psef** tangent bundle, up to a finite étale cover, admits a smooth fibration to an abelian variety whose fibre is rationally connected and has psef tangent bundle.

2. Previous results (Positivity $\Rightarrow$ Geometry)

Theorem (Hosono-Iwai-Matsumura, 2022)

A complex projective manifold with **psef** tangent bundle, up to a finite étale cover, admits a smooth fibration to an abelian variety whose fibre is rationally connected and has psef tangent bundle.

$$\begin{array}{ccc} \tilde{X} & \xrightarrow{\text{alb}_{\tilde{X}}} & A \\ \pi \downarrow & & \\ X & & \end{array}$$

- $\text{alb}_{\tilde{X}}$ is the Albanese morphism of $\tilde{X}$, and π is a finite étale cover.
- A fibre F of $\text{alb}_{\tilde{X}}$ is smooth rationally connected with psef $\mathcal{T}_F$.
- Question (Hosono-Iwai-Matsumura, 2022): Is $-K_F$ big?
- For a quasi-homogeneous rationally connected manifold, (Fu-Zhang, 2016) proved that the anti-canonical divisor is big.

2. Previous results (Dynamics $\Rightarrow$ Geometry)

Conjecture A (folklore conjecture)

A smooth Fano variety of Picard number 1 admitting a non-isomorphic surjective endomorphism is $\mathbb{P}^n$.

2. Previous results (Dynamics $\Rightarrow$ Geometry)

Conjecture A (folklore conjecture)

A smooth Fano variety of Picard number 1 admitting a non-isomorphic surjective endomorphism is $\mathbb{P}^n$.

The following conjectures are natural generalizations of Conjecture A.

Conjecture B (Fakhruddin, 2003); (Meng-Z., 2023)

A smooth rationally connected projective variety admitting an int-amplified endomorphism is toric.

2. Previous results (Dynamics $\Rightarrow$ Geometry)

Conjecture A (folklore conjecture)

A smooth Fano variety of Picard number 1 admitting a non-isomorphic surjective endomorphism is $\mathbb{P}^n$.

The following conjectures are natural generalizations of Conjecture A.

Conjecture B (Fakhruddin, 2003); (Meng-Z., 2023)

A smooth rationally connected projective variety admitting an int-amplified endomorphism is toric.

The most general question in this direction seems to be:

Question C (Meng-Z., 2024); cf. (Hosono-Iwai-Matsumura, 2022)

Let X be a normal $\mathbb{Q}$ -factorial proj. klt variety admitting an int-amplified endomorphism. Then up to an f -equivariant quasi-étale cover, X admits a locally trivial toric fibration over an abelian variety.

2. Previous results (Dynamics $\Rightarrow$ Geometry)

Theorem (Amerik-Rovinski-Van de Ven, 1999), (Hwang-Mok, 2003)

Conjecture A (✓) in dim 3: a smooth projective threefold of Pic #1 w/ a non-isomorphic surjective endomorphism is $\mathbb{P}^3$.

2. Previous results (Dynamics $\Rightarrow$ Geometry)

Theorem (Amerik-Rovinski-Van de Ven, 1999), (Hwang-Mok, 2003)

Conjecture A (✓) in dim 3: a smooth projective threefold of Pic #1 w/ a non-isomorphic surjective endomorphism is $\mathbb{P}^3$.

Theorem (Nakayama, 2002)

Conjecture B and Question C (✓) in dim 2: a smooth rational surface w/ a non-isomorphic surjective endomorphism is toric.

2. Previous results (Dynamics $\Rightarrow$ Geometry)

Theorem (Amerik-Rovinski-Van de Ven, 1999), (Hwang-Mok, 2003)

Conjecture A (✓) in dim 3: a smooth projective threefold of Pic #1 w/ a non-isomorphic surjective endomorphism is $\mathbb{P}^3$.

Theorem (Nakayama, 2002)

Conjecture B and Question C (✓) in dim 2: a smooth rational surface w/ a non-isomorphic surjective endomorphism is toric.

Theorem (Meng-Zhang-Z., 2022)

Conjecture B (✓) Fano threefolds: a smooth Fano threefold is toric if and only if it admits an int-amplified endomorphism. Moreover, a smooth Fano threefold w/ a non-isomorphic endomorphism is rational.

2. Previous results (Dynamics $\Rightarrow$ Geometry)

Theorem (Amerik-Rovinski-Van de Ven, 1999), (Hwang-Mok, 2003)

Conjecture A (✓) in dim 3: a smooth projective threefold of Pic #1 w/ a non-isomorphic surjective endomorphism is $\mathbb{P}^3$.

Theorem (Nakayama, 2002)

Conjecture B and Question C (✓) in dim 2: a smooth rational surface w/ a non-isomorphic surjective endomorphism is toric.

Theorem (Meng-Zhang-Z., 2022)

Conjecture B (✓) Fano threefolds: a smooth Fano threefold is toric if and only if it admits an int-amplified endomorphism. Moreover, a smooth Fano threefold w/ a non-isomorphic endomorphism is rational.

- (Kawakami-Totaro, 2023) and (Totaro, 2023) reproved the above theorems from the viewpoints of Bott vanishing: a normal projective variety w/ an int-amplified endomorphism satisfies Bott vanishing.

2. Previous results (Dynamics $\Rightarrow$ Geometry)

Theorem (Yoshikawa, 2021), (Meng, 2020), (Meng-Zhang, 2018)

Let X be a $\mathbb{Q}$ -factorial proj. klt variety w/ an int-amplified endomorphism. Then there exists an f -equivariant quasi-étale finite cover $\tilde{X} \rightarrow X$ such that $\text{alb}_{\tilde{X}}$ is a fibre space and of Fano type over an abelian variety.

$$\begin{array}{ccc} \tilde{f} \circlearrowleft & \tilde{X} & \xrightarrow{\text{alb}_{\tilde{X}}} A \circlearrowleft f_A \\ & \downarrow \pi & \\ f \circlearrowleft & X & \end{array}$$

2. Previous results (Dynamics $\Rightarrow$ Geometry)

Theorem (Yoshikawa, 2021), (Meng, 2020), (Meng-Zhang, 2018)

Let X be a $\mathbb{Q}$ -factorial proj. klt variety w/ an int-amplified endomorphism. Then there exists an f -equivariant quasi-étale finite cover $\tilde{X} \rightarrow X$ such that $\text{alb}_{\tilde{X}}$ is a fibre space and of Fano type over an abelian variety.

$$\begin{array}{ccc} \tilde{f} \circlearrowleft & \tilde{X} & \xrightarrow{\text{alb}_{\tilde{X}}} A \circlearrowright f_A \\ & \downarrow \pi & \\ f \circlearrowleft & X & \end{array}$$

- When X is smooth, it is not hard to prove that $\text{alb}_{\tilde{X}}$ is smooth.
- To answer Question C, we are left to study the Fano type variety admitting an int-amplified endomorphism (toric?).
- An issue arises after taking π : $\tilde{X}$ may not be $\mathbb{Q}$ -factorial (ex exists?).

2. Previous results (Dynamics $\Rightarrow$ Geometry)

Theorem (Meng-Zhang, 2022)

Let X be a $\mathbb{Q}$ -Gorenstein normal projective variety w/ an int-amplified endomorphism. Then $-K_X \sim_{\mathbb{Q}} D$ for some effective $\mathbb{Q}$ -Cartier divisor D . If $\kappa(X, -K_X) = 0$, then D is reduced such that $g^{-1}(D) = D$ and $g|_{X \setminus D}$ is étale in codimension 1, for any surjective endomorphism g of X .

2. Previous results (Dynamics $\Rightarrow$ Geometry)

Theorem (Meng-Zhang, 2022)

Let X be a $\mathbb{Q}$ -Gorenstein normal projective variety w/ an int-amplified endomorphism. Then $-K_X \sim_{\mathbb{Q}} D$ for some effective $\mathbb{Q}$ -Cartier divisor D . If $\kappa(X, -K_X) = 0$, then D is reduced such that $g^{-1}(D) = D$ and $g|_{X \setminus D}$ is étale in codimension 1, for any surjective endomorphism g of X .

- If $\kappa(X, -K_X) > 0$, (Meng-Zhang, 2022) establishes the equivariant anti-Kodaira fibration to attack several dynamical problems.
- The troublesome case is $\kappa(X, -K_X) = 0$ (EMMP does not work).

2. Previous results (Dynamics $\Rightarrow$ Geometry)

Theorem (Meng-Zhang, 2022)

Let X be a $\mathbb{Q}$ -Gorenstein normal projective variety w/ an int-amplified endomorphism. Then $-K_X \sim_{\mathbb{Q}} D$ for some effective $\mathbb{Q}$ -Cartier divisor D . If $\kappa(X, -K_X) = 0$, then D is reduced such that $g^{-1}(D) = D$ and $g|_{X \setminus D}$ is étale in codimension 1, for any surjective endomorphism g of X .

- If $\kappa(X, -K_X) > 0$, (Meng-Zhang, 2022) establishes the equivariant anti-Kodaira fibration to attack several dynamical problems.
- The troublesome case is $\kappa(X, -K_X) = 0$ (EMMP does not work).
- Our goal here is to understand the explicit geometry of the above case, which we called **the case TIR (totally invariant ramifications)**.
- Clearly, $g^{[*]} \mathcal{T}_X(-\log D) = \mathcal{T}_X(-\log D)$ and $K_X + D \equiv g^*(K_X + D)$.
- When g is int-amplified, we have $K_X + D \equiv 0$ as all eigenvalues of $g^*|_{N^1(X)}$ are of modulus > 1 .

3. Main results (Dynamics $\Rightarrow$ Positivity)

Proposition A (essentially due to (Meng-Z., 2024))

Let $f: X \rightarrow X$ be an int-amplified endomorphism of a normal projective variety X and D a reduced divisor with $f^{-1}(D) = D$. Then the logarithmic tangent sheaf $\mathcal{T}_X(-\log D)$ is psef. Moreover, if X is smooth rationally connected, then $\mathcal{T}_X$ is generically ample.

3. Main results (Dynamics $\Rightarrow$ Positivity)

Proposition A (essentially due to (Meng-Z., 2024))

Let $f: X \rightarrow X$ be an int-amplified endomorphism of a normal projective variety X and D a reduced divisor with $f^{-1}(D) = D$. Then the logarithmic tangent sheaf $\mathcal{T}_X(-\log D)$ is psef. Moreover, if X is smooth rationally connected, then $\mathcal{T}_X$ is generically ample.

- It relates (Hosono-Iwai-Matsumura, 2022) to (Yoshikawa, 2021).
- If we take X to be a smooth hypersurface and $D = 0$, then together with (Höring-Liu-Shao, 2022), we give another proof of (Beauville, 2001): $\deg(X) \leq 2$. By (Paranjape and Srinivas, 1989), $\deg(X) = 1$.
- For a Fano mfd X of $\rho(X) = 1$ w/ a non-isom. surj. endo., if $\mathcal{T}_X$ is big and the VMRT along a general point is not dual defective, then $X = \mathbb{P}^n$ (Shao-Z., 2024). Theorem A gives a step to the expectation:

Int-amplified endomorphisms & Smooth rationally connected $\Rightarrow$ big $\mathcal{T}_X$

3. Main results (Proof of Theorem A)

Proposition A (essentially due to (Meng-Z., 2024))

Let $f: X \rightarrow X$ be an int-amplified endomorphism of a normal projective variety X and D a reduced divisor with $f^{-1}(D) = D$. Then the logarithmic tangent sheaf $\mathcal{T}_X(-\log D)$ is psef.

We briefly explain the idea of the proof (for simplicity, taking $D = 0$):

3. Main results (Proof of Theorem A)

Proposition A (essentially due to (Meng-Z., 2024))

Let $f: X \rightarrow X$ be an int-amplified endomorphism of a normal projective variety X and D a reduced divisor with $f^{-1}(D) = D$. Then the logarithmic tangent sheaf $\mathcal{T}_X(-\log D)$ is psef.

We briefly explain the idea of the proof (for simplicity, taking $D = 0$):

- Use the injection $0 \rightarrow \mathcal{T}_X \rightarrow (f^m)^{[*]} \mathcal{T}_X$ for any m ;

3. Main results (Proof of Theorem A)

Proposition A (essentially due to (Meng-Z., 2024))

Let $f: X \rightarrow X$ be an int-amplified endomorphism of a normal projective variety X and D a reduced divisor with $f^{-1}(D) = D$. Then the logarithmic tangent sheaf $\mathcal{T}_X(-\log D)$ is psef.

We briefly explain the idea of the proof (for simplicity, taking $D = 0$):

- Use the injection $0 \rightarrow \mathcal{T}_X \rightarrow (f^m)^{[*]} \mathcal{T}_X$ for any m ;
- $\mathrm{Sym}^{[s]} \mathcal{T}_X \otimes \mathcal{O}_X(H)$ is psef for any s : indeed, tensoring a sufficiently large ample line bundle outside is "equivalent" to tensoring a rather small multiple line bundle inside $(f^m)^*$ if we take $m \gg 1$.

3. Main results (Proof of Theorem A)

Proposition A (essentially due to (Meng-Z., 2024))

Let $f: X \rightarrow X$ be an int-amplified endomorphism of a normal projective variety X and D a reduced divisor with $f^{-1}(D) = D$. Then the logarithmic tangent sheaf $\mathcal{T}_X(-\log D)$ is psef.

We briefly explain the idea of the proof (for simplicity, taking $D = 0$):

- Use the injection $0 \rightarrow \mathcal{T}_X \rightarrow (f^m)^{[*]} \mathcal{T}_X$ for any m ;
- $\mathrm{Sym}^{[s]} \mathcal{T}_X \otimes \mathcal{O}_X(H)$ is psef for any s : indeed, tensoring a sufficiently large ample line bundle outside is "equivalent" to tensoring a rather small multiple line bundle inside $(f^m)^*$ if we take $m \gg 1$.
- "closedness" implies $\mathrm{Sym}^{[s]} \mathcal{T}_X$ itself is psef;

3. Main results (Proof of Theorem A)

Proposition A (essentially due to (Meng-Z., 2024))

Let $f: X \rightarrow X$ be an int-amplified endomorphism of a normal projective variety X and D a reduced divisor with $f^{-1}(D) = D$. Then the logarithmic tangent sheaf $\mathcal{T}_X(-\log D)$ is psef.

We briefly explain the idea of the proof (for simplicity, taking $D = 0$):

- Use the injection $0 \rightarrow \mathcal{T}_X \rightarrow (f^m)^{[*]} \mathcal{T}_X$ for any m ;
- $\mathrm{Sym}^{[s]} \mathcal{T}_X \otimes \mathcal{O}_X(H)$ is psef for any s : indeed, tensoring a sufficiently large ample line bundle outside is "equivalent" to tensoring a rather small multiple line bundle inside $(f^m)^*$ if we take $m \gg 1$.
- "closedness" implies $\mathrm{Sym}^{[s]} \mathcal{T}_X$ itself is psef;
- Suppose X is smooth rationally connected but $\mathcal{T}_X$ is not generically ample. Then there exists a quotient bundle (still being psef) with trivial determinant. Hence, it is a direct sum of trivial line bundle and it implies that Ω_X contains a section, a contradiction.

3. Main results (Dynamics $\Rightarrow$ Positivity)

Let $\text{SEnd}(X, D)$ be the monoid of surj. endo. f of X with $f^{-1}(D) = D$. The following theorem is an extremal case of Proposition A when the divisor D being maximal.

Theorem B (Equivariant toric cover) (Meng-Z., 2024)

Let X be a projective klt variety and $D \geq 0$ a reduced divisor. Assume that $K_X + D \equiv 0$ and $\text{SEnd}(X, D)$ contains an int-amplified endo. f . Then $\mathcal{T}_X(-\log D)$ is numerically flat.

3. Main results (Dynamics $\Rightarrow$ Positivity)

Let $\text{SEnd}(X, D)$ be the monoid of surj. endo. f of X with $f^{-1}(D) = D$. The following theorem is an extremal case of Proposition A when the divisor D being maximal.

Theorem B (Equivariant toric cover) (Meng-Z., 2024)

Let X be a projective klt variety and $D \geq 0$ a reduced divisor. Assume that $K_X + D \equiv 0$ and $\text{SEnd}(X, D)$ contains an int-amplified endo. f . Then $\mathcal{T}_X(-\log D)$ is numerically flat.

Moreover, $\forall g \in \text{SEnd}(X, D)$, $\exists$ a quasi-étale cover $\pi : \tilde{X} \rightarrow X$ satisfying:

- The pair $(\tilde{X}, \pi^*D)$ admits a Zariski locally trivial toric fibration over an abelian variety A ,
 - there exists $\tilde{g} \in \text{SEnd}(\tilde{X}, \pi^*D)$ such that $\pi \circ \tilde{g} = g^s \circ \pi$ for some $s > 0$,
 - $\text{SEnd}(\tilde{X}, \pi^*D)$ contains an int-amplified endomorphism (but it may not be the lifting of f !).
-
- The condition $K_X + D \equiv 0 \Leftrightarrow f^{-1}(D) = D$ and $f|_{X \setminus D}$ is quasi-étale, i.e., $K_X + D = f^*(K_X + D)$.

3. Main results (Dynamics $\Rightarrow$ Positivity)

- When X is a Fano manifold of Picard number 1, (Hwang-Nakayama, 2011) proved that X is the projective space.
- When X is smooth rationally connected, (Meng-Zhang, 2019) and (Meng-Z., 2023) proved that X is a toric variety.
- When X is of Fano type: (Moraga-Yanez-Yeong, 2024) proved that up to a finite cover, X is toric (a completely different approach).

3. Main results (Dynamics $\Rightarrow$ Positivity)

- When X is a Fano manifold of Picard number 1, (Hwang-Nakayama, 2011) proved that X is the projective space.
- When X is smooth rationally connected, (Meng-Zhang, 2019) and (Meng-Z., 2023) proved that X is a toric variety.
- When X is of Fano type: (Moraga-Yanez-Yeong, 2024) proved that up to a finite cover, X is toric (a completely different approach).
- Be careful: the upstairs is not necessarily $\mathbb{Q}$ -factorial anymore. In certain good case, we can take a higher smooth model:

3. Main results (Dynamics $\Rightarrow$ Positivity)

- When X is a Fano manifold of Picard number 1, (Hwang-Nakayama, 2011) proved that X is the projective space.
- When X is smooth rationally connected, (Meng-Zhang, 2019) and (Meng-Z., 2023) proved that X is a toric variety.
- When X is of Fano type: (Moraga-Yanez-Yeong, 2024) proved that up to a finite cover, X is toric (a completely different approach).
- Be careful: the upstairs is not necessarily $\mathbb{Q}$ -factorial anymore. In certain good case, we can take a higher smooth model:

Theorem C (Equivariant resolution) (Meng-Z., 2024)

Let $\tilde{X}$ be a normal projective variety and $\tilde{D} \geq 0$ a reduced divisor. Let $\pi: (\tilde{X}, \tilde{D}) \rightarrow A$ be a Zariski locally trivial toric fibration over an abelian variety A . Then there exists a resolution $\sigma: \hat{X} \rightarrow \tilde{X}$ such that for any $\tilde{f} \in \text{SEnd}(\tilde{X}, \tilde{D})$ with $\tilde{f}^* \tilde{D} = q \tilde{D}$, there exists some $\hat{f} \in \text{SEnd}(\hat{X}, \sigma^{-1}(\tilde{D}))$ with $\sigma \circ \hat{f} = \tilde{f}^s \circ \sigma$ for some $s > 0$.

3. Main results (Proof of Theorem B)

- By Proposition A, $\mathcal{T}_X(-\log D)$ is psef.
- The numerical flatness follows from $K_X + D \equiv 0$ and the following:

Proposition (Iwai-Matsumura-Z., 2023), (Meng-Z., 2024)

Let X be a projective klt variety over $k = \bar{k}$ of char $k = 0$, and $\mathcal{E}$ a psef reflexive sheaf on X . If $\det(\mathcal{E}) \cdot H^{\dim(X)-1} = 0$ for an ample divisor H , then $\exists$ a quasi-étale cover s.t. the reflexive pullback of $\mathcal{E}$ is numerically flat.

- The above proposition extends (Cao-Höring, 2019), (Höring-Peternell, 2019), and (Hosono-Iwai-Matsumura, 2022).

3. Main results (Proof of Theorem B)

- By Proposition A, $\mathcal{T}_X(-\log D)$ is psef.
- The numerical flatness follows from $K_X + D \equiv 0$ and the following:

Proposition (Iwai-Matsumura-Z., 2023), (Meng-Z., 2024)

Let X be a projective klt variety over $k = \bar{k}$ of char $k = 0$, and $\mathcal{E}$ a psef reflexive sheaf on X . If $\det(\mathcal{E}) \cdot H^{\dim(X)-1} = 0$ for an ample divisor H , then $\exists$ a quasi-étale cover s.t. the reflexive pullback of $\mathcal{E}$ is numerically flat.

- The above proposition extends (Cao-Höring, 2019), (Höring-Peternell, 2019), and (Hosono-Iwai-Matsumura, 2022).
- It has the own interest to deduce the structure theorems for projective klt varieties w/ certain positivity. (Matsumura, 2023) established the equivariant minimal model program for projective klt varieties with psef tangent sheaf and the above proposition characterized the end product: a quasi-étale quotient of an abelian variety.

3. Main results (Proof of Theorem B)

- By Proposition A, $\mathcal{T}_X(-\log D)$ is psef.
- The numerical flatness follows from $K_X + D \equiv 0$ and the following:

Proposition (Iwai-Matsumura-Z., 2023), (Meng-Z., 2024)

Let X be a projective klt variety over $k = \bar{k}$ of char $k = 0$, and $\mathcal{E}$ a psef reflexive sheaf on X . If $\det(\mathcal{E}) \cdot H^{\dim(X)-1} = 0$ for an ample divisor H , then $\exists$ a quasi-étale cover s.t. the reflexive pullback of $\mathcal{E}$ is numerically flat.

- The above proposition extends (Cao-Höring, 2019), (Höring-Peternell, 2019), and (Hosono-Iwai-Matsumura, 2022).
- It has the own interest to deduce the structure theorems for projective klt varieties w/ certain positivity. (Matsumura, 2023) established the equivariant minimal model program for projective klt varieties with psef tangent sheaf and the above proposition characterized the end product: a quasi-étale quotient of an abelian variety.
- As $\mathcal{T}_X(-\log D)$ is numerically flat, we can conclude the structure essentially by (Druel-Lo Bianco, 2022).

3. Main results (Proof of Theorem B)

Theorem (Druel-Lo Bianco, 2022), (Meng-Z., 2024)

Let X be a projective klt variety over $k = \bar{k}$ of char $k = 0$. Let $D \geq 0$ be a reduced divisor s.t. $\Omega_X^{[1]}(\log D)$ is numerically flat. Then the augmented irregularity $\tilde{q}(X) < \infty$ holds. Suppose further that (X, D) is lc with $q(X) = \tilde{q}(X)$. Then the albanese morphism $(X, D) \rightarrow A$ is an analytically locally trivial toric fibration.

3. Main results (Proof of Theorem B)

Theorem (Druel-Lo Bianco, 2022), (Meng-Z., 2024)

Let X be a projective klt variety over $k = \bar{k}$ of char $k = 0$. Let $D \geq 0$ be a reduced divisor s.t. $\Omega_X^{[1]}(\log D)$ is numerically flat. Then the augmented irregularity $\tilde{q}(X) < \infty$ holds. Suppose further that (X, D) is lc with $q(X) = \tilde{q}(X)$. Then the albanese morphism $(X, D) \rightarrow A$ is an analytically locally trivial toric fibration.

- (Druel-Lo Bianco, 22) is formulated in terms of MRC fibrations over $\mathbb{C}$. By the transcendental argument, one needs to be careful for the extension from the MRC over $\mathbb{C}$ to the MRC over a countable field (e.g., certain K3 surfaces over $\overline{\mathbb{Q}}$).
- Alb however is independent of the choice of k .
- Use a technical factorization of D , we even make the toric fibration Zariski locally trivial after a further cover $\Leftrightarrow$ the restriction of each irreducible component of D to a fibre remaining irreducible.
- Dynamical liftings of f and g are more technical, which we omit.

4. Application: the Kawaguchi-Silverman conjecture

We shall give an affirmative answer to certain arithmetic conjectures for the case of int-amplified endomorphisms:

Corollary (Meng-Z., 2024)

Let X be a $\mathbb{Q}$ -factorial projective klt variety admitting an int-amplified endomorphism. Then both the Kawaguchi-Silverman conjecture (KSC) and the small Arithmetic Non-Density (sAND) conjecture hold for any surjective endomorphism of X .

4. Application: the Kawaguchi-Silverman conjecture

We shall give an affirmative answer to certain arithmetic conjectures for the case of int-amplified endomorphisms:

Corollary (Meng-Z., 2024)

Let X be a $\mathbb{Q}$ -factorial projective klt variety admitting an int-amplified endomorphism. Then both the Kawaguchi-Silverman conjecture (KSC) and the small Arithmetic Non-Density (sAND) conjecture hold for any surjective endomorphism of X .

We focus on the KSC below. Let f be a surjective endomorphism of a projective variety X over $\overline{\mathbb{Q}}$, and H an ample divisor on X . There are two fundamental dynamical invariants.

4. Application: the Kawaguchi-Silverman conjecture

We shall give an affirmative answer to certain arithmetic conjectures for the case of int-amplified endomorphisms:

Corollary (Meng-Z., 2024)

Let X be a $\mathbb{Q}$ -factorial projective klt variety admitting an int-amplified endomorphism. Then both the Kawaguchi-Silverman conjecture (KSC) and the small Arithmetic Non-Density (sAND) conjecture hold for any surjective endomorphism of X .

We focus on the KSC below. Let f be a surjective endomorphism of a projective variety X over $\overline{\mathbb{Q}}$, and H an ample divisor on X . There are two fundamental dynamical invariants.

- The first dynamical degree is defined as

$$\delta_f := \lim_{m \rightarrow \infty} ((f^m)^* H \cdot H^{\dim(X)-1})^{1/m}.$$

Such limit, independent of the choice of H , exists and equals to the spectral radius of $f^*|_{N^1(X)}$ (Dinh-Sibony).

4. Application: the Kawaguchi-Silverman conjecture

- The arithmetic degree of f at $x \in X(\overline{\mathbb{Q}})$ is defined as

$$\alpha_f(x) := \lim_{m \rightarrow \infty} \max\{1, h_H(f^m(x))\}^{1/m},$$

where h_H is the Weil height function associated with H . Such limit exists and is independent of the choice of H (Kawaguchi-Silverman).

4. Application: the Kawaguchi-Silverman conjecture

- The arithmetic degree of f at $x \in X(\overline{\mathbb{Q}})$ is defined as

$$\alpha_f(x) := \lim_{m \rightarrow \infty} \max\{1, h_H(f^m(x))\}^{1/m},$$

where h_H is the Weil height function associated with H . Such limit exists and is independent of the choice of H (Kawaguchi-Silverman).

- (Kawaguchi-Silverman, 2016) proves $\alpha_f(x) \leq \delta_f$, i.e., the arithmetic complexity of the forward orbit of x does not exceed the geometric complexity of iterations of f .

4. Application: the Kawaguchi-Silverman conjecture

- The arithmetic degree of f at $x \in X(\overline{\mathbb{Q}})$ is defined as

$$\alpha_f(x) := \lim_{m \rightarrow \infty} \max\{1, h_H(f^m(x))\}^{1/m},$$

where h_H is the Weil height function associated with H . Such limit exists and is independent of the choice of H (Kawaguchi-Silverman).

- (Kawaguchi-Silverman, 2016) proves $\alpha_f(x) \leq \delta_f$, i.e., the arithmetic complexity of the forward orbit of x does not exceed the geometric complexity of iterations of f .
- (Kawaguchi-Silverman, 2016) conjectures that if the forward orbit of x is Zariski dense in $X(\overline{\mathbb{Q}})$, then $\alpha_f(x) = \delta_f$.

4. Application: the Kawaguchi-Silverman conjecture

- The arithmetic degree of f at $x \in X(\overline{\mathbb{Q}})$ is defined as

$$\alpha_f(x) := \lim_{m \rightarrow \infty} \max\{1, h_H(f^m(x))\}^{1/m},$$

where h_H is the Weil height function associated with H . Such limit exists and is independent of the choice of H (Kawaguchi-Silverman).

- (Kawaguchi-Silverman, 2016) proves $\alpha_f(x) \leq \delta_f$, i.e., the arithmetic complexity of the forward orbit of x does not exceed the geometric complexity of iterations of f .
- (Kawaguchi-Silverman, 2016) conjectures that if the forward orbit of x is Zariski dense in $X(\overline{\mathbb{Q}})$, then $\alpha_f(x) = \delta_f$.
- (✓) polarized endo. (Kawaguchi-Silverman, 2016); abelian varieties (Kawaguchi-Silverman, 2016); $\dim = 2$ (Matsuzawa-Sano-Shibata, 2018), (Meng-Zhang, 2022); Mori dream space (Matsuzawa, 2020); HyperKähler varieties (Lesieutre-Satriano, 2021); Smooth threefolds with $\deg(f) > 1$ (Lesieutre-Satriano, 2021), (Meng-Zhang, 2023) etc.

4. Application: the Kawaguchi-Silverman conjecture

Theorem (Meng-Zhang, 22), (Matsuzawa-Meng-Shibata-Zhang, 23)

Let X be a $\mathbb{Q}$ -factorial klt projective variety admitting an int-amplified endomorphism. Then we have:

- (1) If K_X is psef, then KSC/sAND (✓) for any surjective endomorphism of X .
- (2) Suppose that KSC/sAND holds for the Case TIR. Then KSC/sAND (✓) for any surjective endomorphism g of X .

4. Application: the Kawaguchi-Silverman conjecture

Theorem (Meng-Zhang, 22), (Matsuzawa-Meng-Shibata-Zhang, 23)

Let X be a $\mathbb{Q}$ -factorial klt projective variety admitting an int-amplified endomorphism. Then we have:

- (1) If K_X is psef, then KSC/sAND (✓) for any surjective endomorphism of X .
- (2) Suppose that KSC/sAND holds for the Case TIR. Then KSC/sAND (✓) for any surjective endomorphism g of X .

Let X be a $\mathbb{Q}$ -factorial projective klt variety admitting an int-amplified endomorphism, and $g: X \rightarrow X$ any surjective endomorphism. By our structure theorem and running the equivariant MMP, we are reduced to:

- (A1) The anti-Kodaira dimension $\kappa(X, -K_X) = 0$; $-K_X$ is nef, whose class is extremal in the nef cone and the psef cone.
- (A2) $f^*D = \delta_f D$ for some prime divisor $D \sim_{\mathbb{Q}} -K_X$.
- (A3) The ramification divisor of f satisfies $\text{Supp } R_f = D$. **Contradiction!**

Thank you for your attention!